



Hospital Engineering and Management

Lecture 3: Outpatient Services and Patient-Flow Planning

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Course Code: UBM702

Learning Objectives

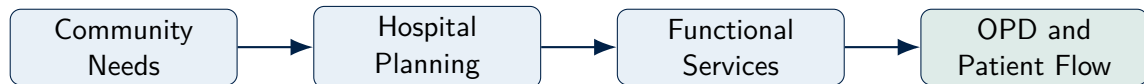
By the end of this lecture, students should be able to:

- ▶ explain the purpose and scope of outpatient services in a hospital;
- ▶ describe the major functional areas of an outpatient department;
- ▶ map the patient journey from arrival to exit;
- ▶ apply basic patient-flow and queue-management principles;
- ▶ identify engineering, safety and infection-control requirements of OPD areas;
- ▶ explain the role of biomedical engineers in OPD planning and operation.

Connection with the Previous Lecture

Lecture 2 Established

Hospital planning translates healthcare needs into service capacity, space, zoning, circulation, organization and utility requirements.



Focus of Lecture 3

We now study the outpatient department as the main entry point for planned patient care.

Opening Question: Why Does OPD Matter?

Think About It

A patient visits a hospital for fever, diabetes follow-up or chest pain screening. Which hospital systems must work before the patient even meets the doctor?

- ▶ registration and appointment system;
- ▶ waiting and information desk;
- ▶ consultation rooms;
- ▶ sample collection and laboratory;
- ▶ radiology and diagnostic services;
- ▶ pharmacy and billing;
- ▶ medical records and HMIS;
- ▶ safe circulation and infection control.

For many patients, OPD is the first impression of hospital quality.

What Are Outpatient Services?

Definition

Outpatient services provide consultation, diagnosis, minor procedures, preventive care and follow-up treatment without admitting the patient to an inpatient bed.

- ▶ Patients enter, receive service and usually leave on the same day.
- ▶ OPD reduces unnecessary admissions and supports continuity of care.
- ▶ It connects clinical care with laboratories, radiology, pharmacy and records.
- ▶ Efficient OPD design reduces waiting time, crowding and patient dissatisfaction.

$$\text{OPD} = \text{Access} + \text{Consultation} + \text{Diagnosis} + \text{Follow-up}$$

Role of OPD in the Hospital System

- ▶ first contact point for planned care;
- ▶ screening and referral to specialists;
- ▶ chronic-disease follow-up;
- ▶ preventive and community-health services;
- ▶ pre-admission evaluation;
- ▶ post-discharge review.

Management View

OPD workload strongly affects registration load, diagnostic demand, pharmacy workload, records management and hospital reputation.

Engineering View

OPD requires reliable equipment, IT systems, electrical supply, ventilation, lighting, signage and patient-safe movement.

Common Types of Outpatient Services

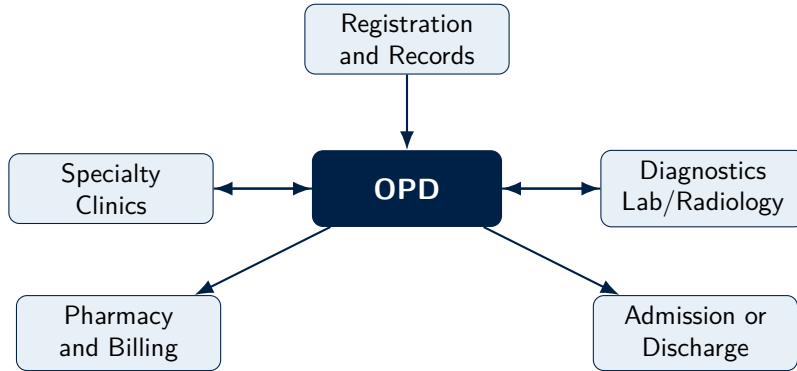
General OPD

- ▶ medicine;
- ▶ surgery;
- ▶ obstetrics and gynecology;
- ▶ pediatrics;
- ▶ orthopedics;
- ▶ dermatology and ENT.

Special and Support Clinics

- ▶ cardiology and neurology clinic;
- ▶ diabetes and hypertension clinic;
- ▶ physiotherapy and rehabilitation;
- ▶ vaccination clinic;
- ▶ minor procedure room;
- ▶ teleconsultation unit.

OPD as the Front Door of the Hospital



Key Idea

OPD performance depends on many linked departments; a delay in one unit can increase crowding in the whole outpatient area.

Typical Patient Journey in OPD



Design Challenge

The actual path may include billing, sample collection, imaging, minor procedures, admission decision or referral. Good planning must handle both simple and complex paths.

Planning Principles for OPD

- ▶ locate OPD near the main entrance and public transport access;
- ▶ provide clear separation between outpatient, emergency and inpatient circulation;
- ▶ keep registration, waiting, consultation, diagnostics and pharmacy logically connected;
- ▶ provide sufficient waiting space, toilets, drinking water and information support;
- ▶ ensure barrier-free access for elderly and differently abled patients;
- ▶ keep noisy, crowded and infection-risk areas away from sterile and critical zones;
- ▶ plan for future growth in patient volume and diagnostic demand.

OPD planning is patient-flow planning.

Workload and Capacity Planning

OPD capacity depends on expected patient load, consultation time, clinic hours and staffing.

Important Inputs

- ▶ average OPD visits per day;
- ▶ peak-hour patient arrival rate;
- ▶ average consultation time;
- ▶ number of specialties;
- ▶ working hours and doctor availability;
- ▶ diagnostic and pharmacy demand.

Planning Question

If one doctor can consult 6 patients per hour, how many consultation rooms are needed for a peak demand of 42 patients per hour?

$$\frac{42}{6} = 7 \text{ rooms}$$

Simple Capacity Estimation

Consultation-Room Requirement

$$N_c = \left\lceil \frac{P_d \times t_c}{H \times 60 \times \eta} \right\rceil$$

- ▶ N_c : number of consultation rooms;
- ▶ P_d : expected outpatient visits per day;
- ▶ t_c : average consultation time in minutes;
- ▶ H : available consultation hours per day;
- ▶ η : practical utilization factor, normally less than 1.

Interpretation

Capacity should be based on peak demand and practical delays, not only daily averages.

Example: Consultation-Room Estimation

A hospital expects 480 OPD visits per day. Average consultation time is 8 minutes. OPD runs for 8 hours. Assume practical utilization $\eta = 0.80$.

Calculation

$$N_c = \left\lceil \frac{480 \times 8}{8 \times 60 \times 0.80} \right\rceil = \left\lceil \frac{3840}{384} \right\rceil = 10$$

Planning Result

At least 10 consultation rooms are required under these assumptions.

Practical Note

More rooms may be required when specialties, peak-hour arrivals or follow-up reviews are considered.

Registration and Appointment System

- ▶ patient identification and demographic data;
- ▶ appointment scheduling;
- ▶ token or queue number generation;
- ▶ billing and insurance verification;
- ▶ retrieval or creation of medical record;
- ▶ guidance to clinic, lab or diagnostic area.

Design Requirements

- ▶ counters visible from entrance;
- ▶ adequate queuing space;
- ▶ digital display and public-address support;
- ▶ separate help desk for elderly or critical patients;
- ▶ data privacy at counters.

Waiting Area Design

Waiting spaces affect comfort, safety and perception of service quality.

- ▶ adequate seating capacity;
- ▶ natural or mechanical ventilation;
- ▶ clear signage and queue display;
- ▶ drinking water and toilets;
- ▶ wheelchair movement space;
- ▶ noise control and lighting;
- ▶ isolation possibility for infectious symptoms.

Patient-Centered Question

Can a first-time patient understand where to go without repeatedly asking staff?

Engineering Concern

Overcrowded waiting areas increase thermal load, ventilation demand, infection risk and evacuation difficulty.

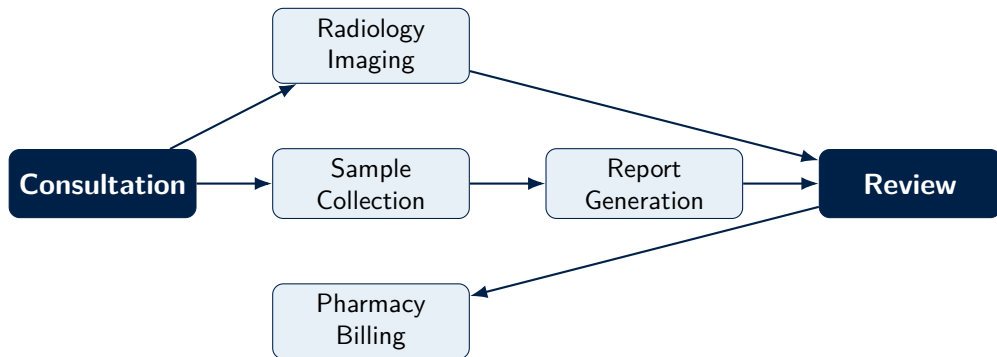
Consultation and Examination Rooms

Typical Functions

Clinical history, physical examination, diagnosis, prescription, counselling and referral.

- ▶ doctor work desk;
- ▶ examination couch;
- ▶ hand-washing facility or sanitizer;
- ▶ medical records computer terminal;
- ▶ privacy curtain or partition;
- ▶ basic diagnostic instruments.
- ▶ adequate lighting;
- ▶ electrical sockets with safety earthing;
- ▶ network connectivity;
- ▶ biomedical-waste bins where needed;
- ▶ acoustic privacy;
- ▶ emergency call support in selected rooms.

Diagnostic Linkages from OPD



Planning Implication

Laboratory sample collection, radiology, billing and pharmacy should be accessible without forcing unnecessary crossing of patient, staff and supply routes.

Specialty Clinics and Procedure Areas

- ▶ Specialty clinics may require dedicated examination rooms and diagnostic equipment.
- ▶ Minor procedure rooms require higher infection-control, lighting and waste-management standards.
- ▶ Physiotherapy and rehabilitation areas require open movement space and patient safety rails.
- ▶ Ophthalmology, ENT and dental clinics require equipment-specific electrical and plumbing support.
- ▶ Cardiology OPD may require ECG, echo and emergency response support.
- ▶ Diabetes clinics may require counselling, point-of-care testing and patient-education space.

Biomedical-Engineering View

Equipment planning must consider power rating, calibration, maintenance access, consumables, user training and safety testing.

Patient Flow and Departmental Zoning

Public Zone

entrance, registration, waiting, help desk, toilets and pharmacy access.

Staff and Service Zone

doctor rooms, nurse station, clean utility, dirty utility and stores.

Clinical Zone

consultation rooms, examination rooms, sample collection and minor procedure rooms.

Restricted Zone

areas requiring controlled access, privacy, infection control or radiation safety.

Public flow should be simple; service flow should be controlled.

Accessibility and Patient-Centered Design

OPD users include children, elderly patients, pregnant women, persons with disabilities and patients with pain or anxiety.

- ▶ ramps, lifts and wheelchair-friendly corridors;
- ▶ non-slippery flooring and handrails where needed;
- ▶ readable signage using symbols and local language;
- ▶ seating for elderly, pregnant and disabled patients;
- ▶ privacy during consultation, billing and counselling;
- ▶ drinking water, toilets and mother-child support spaces;
- ▶ feedback mechanism and grievance-support desk.

Design Test

Can a wheelchair user move from entrance to consultation, diagnostics, pharmacy and exit independently?

Infection Control in OPD

Outpatient areas often handle undiagnosed infectious patients.

- ▶ hand hygiene at entry and clinical points;
- ▶ respiratory hygiene and masking support;
- ▶ triage or fast-track for fever and cough;
- ▶ adequate ventilation in crowded waiting areas;
- ▶ cleaning schedule for high-touch surfaces;
- ▶ color-coded waste segregation.

Planning Principle

Infection control is easier when patient flow, ventilation, surface cleaning and waste disposal are planned together.

Examples

separate fever clinic, dedicated sample-collection queue and covered biomedical-waste bins.

Safety and Emergency Readiness in OPD

- ▶ fire exits and evacuation routes should remain visible and unobstructed;
- ▶ crowding should not block corridors, stairs or emergency doors;
- ▶ electrical sockets and equipment cables should not create trip hazards;
- ▶ emergency trolley, oxygen and suction may be needed in selected high-risk areas;
- ▶ security staff should manage crowd control and violent incidents;
- ▶ code-blue or emergency-call protocol should be known to OPD staff;
- ▶ clear communication is required during fire, disaster or mass-casualty events.

Key Message

OPD is not a low-risk area; it handles large crowds and may receive sudden emergencies.

Digital Systems in OPD

Common Digital Tools

- ▶ hospital management information system;
- ▶ electronic medical records;
- ▶ appointment and queue management;
- ▶ laboratory and radiology information systems;
- ▶ e-prescription and pharmacy billing;
- ▶ telemedicine and patient portal.

Benefits

- ▶ shorter queues;
- ▶ reduced duplicate records;
- ▶ faster report access;
- ▶ better patient tracking;
- ▶ improved data for planning.

Engineering Concern

Digital OPD needs reliable network, power backup, data privacy and staff training.

Role of Biomedical Engineers in OPD

Biomedical engineers contribute to OPD planning and operation by supporting:

- ▶ selection and installation of diagnostic and therapeutic equipment;
- ▶ electrical safety testing and equipment calibration;
- ▶ preventive-maintenance schedules;
- ▶ user training for nurses, technicians and doctors;
- ▶ equipment inventory, uptime tracking and spare-parts planning;
- ▶ integration of devices with digital records where applicable;
- ▶ risk assessment for equipment failure, misuse and infection transmission.

In OPD, biomedical engineering supports both patient safety and service speed.

Common Equipment in OPD Areas

General OPD

- ▶ BP apparatus and pulse oximeter;
- ▶ weighing scale and thermometer;
- ▶ examination light;
- ▶ ECG machine;
- ▶ nebulizer and glucometer;
- ▶ basic emergency equipment.

Specialty OPD

- ▶ ultrasound or echo system;
- ▶ dental chair and compressor;
- ▶ ENT microscope or endoscope;
- ▶ slit lamp;
- ▶ physiotherapy equipment;
- ▶ point-of-care analyzers.

Maintenance and Operational Issues

- ▶ high patient turnover increases wear and tear of equipment and furniture;
- ▶ peak-hour crowding increases load on HVAC, toilets, lifts and IT systems;
- ▶ failure of token display, printers or network may disrupt clinical workflow;
- ▶ improper electrical extension boards can create safety hazards;
- ▶ poor calibration may lead to wrong readings and clinical decisions;
- ▶ inadequate cleaning of reusable devices may increase infection risk;
- ▶ absence of backup devices can delay high-volume clinics.

Management Requirement

OPD maintenance should be planned outside peak hours wherever possible.

Performance Indicators for OPD

Patient-Flow Indicators

- ▶ average waiting time;
- ▶ consultation time;
- ▶ total visit duration;
- ▶ number of patients per doctor;
- ▶ no-show rate;
- ▶ revisit and referral rate.

Engineering Indicators

- ▶ equipment uptime;
- ▶ response time to breakdown;
- ▶ calibration compliance;
- ▶ power and network availability;
- ▶ HVAC comfort complaints;
- ▶ safety incidents.

What gets measured can be improved.

Common OPD Design Mistakes

- ▶ locating OPD deep inside the hospital, causing unnecessary movement;
- ▶ insufficient registration counters and waiting space;
- ▶ mixing emergency, outpatient, inpatient and service flows;
- ▶ poor signage and unclear patient navigation;
- ▶ inadequate toilets, drinking water and seating;
- ▶ no isolation pathway for infectious symptoms;
- ▶ lack of maintenance access to equipment and IT systems;
- ▶ failure to plan for future patient-load growth.

Result

These mistakes increase waiting time, crowding, staff stress, infection risk and patient dissatisfaction.

Case Study: Overcrowded OPD

A 300-bed hospital receives nearly 900 outpatient visits per day. Registration begins at 8:30 AM, but many patients arrive by 8:00 AM. Long queues form near the entrance and pharmacy. Doctors complain that reports are delayed.

Discussion Questions

1. Where are the possible bottlenecks in this OPD system?
2. Which areas need more space or better scheduling?
3. How can digital queue management help?
4. Which engineering services may be overloaded?
5. What immediate and long-term improvements can be proposed?

Engineering Analysis of the Case

Possible Bottlenecks

- ▶ registration counters;
- ▶ waiting area;
- ▶ sample collection;
- ▶ radiology reporting;
- ▶ billing and pharmacy;
- ▶ lift and corridor capacity.

Possible Interventions

- ▶ appointment slots;
- ▶ separate follow-up queue;
- ▶ digital token display;
- ▶ decentralized sample collection;
- ▶ added pharmacy counters;
- ▶ real-time monitoring dashboard.

Classroom Activity

Activity: Draw an OPD Flow Map

Work in small groups. Design the patient-flow map for a medium-size hospital OPD.

Your diagram should include:

1. entry and registration;
2. waiting and consultation rooms;
3. laboratory sample collection;
4. radiology access;
5. billing and pharmacy;
6. exit or admission path;
7. one infection-control provision;
8. one biomedical-engineering support point.

Mini Project Ideas Related to OPD

- ▶ OPD patient-flow audit using observation and simple time measurements;
- ▶ queue-management improvement proposal for registration or pharmacy;
- ▶ equipment inventory and maintenance plan for a specialty clinic;
- ▶ accessibility audit of an outpatient area;
- ▶ infection-control audit of waiting and consultation spaces;
- ▶ digital dashboard concept for OPD workload and equipment uptime;
- ▶ layout redesign for reducing patient movement between OPD, laboratory and pharmacy.

Expected Output

Identify one problem, collect basic observations, propose an engineering or management solution.

Summary

- ▶ OPD is the main entry point for planned hospital care.
- ▶ Outpatient services include consultation, diagnosis, follow-up, prevention and minor procedures.
- ▶ OPD planning must integrate registration, waiting, clinics, diagnostics, pharmacy and records.
- ▶ Patient-flow planning reduces waiting, crowding, infection risk and confusion.
- ▶ Biomedical engineers support equipment safety, uptime, calibration and digital integration.
- ▶ Good OPD performance improves patient satisfaction and hospital efficiency.

An efficient OPD converts hospital access into organized patient care.

Assignment for the Next Lecture

Visit or study the OPD layout of any hospital through observation, website information or published layout images.

Prepare a one-page note containing:

1. patient-flow diagram from entry to exit;
2. major OPD service points;
3. one possible bottleneck;
4. one infection-control observation;
5. three important biomedical equipment items;
6. one practical improvement suggestion.

Submission Format

One page, individually prepared, with a labelled flow diagram.

Suggested Reading

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ S. Goel et al., *Textbook of Hospital Administration: A Problem-Solving Approach*, Elsevier India.

Next Lecture: Nursing Unit and Intensive Care Unit Planning

Thank You

Questions and Discussion